



3252 - New Predictions for Galaxy Metallicity Scaling Relations in the JWST Era: Departures from Equilibrium Models and Implications for the Cosmic Baryon Cycle

Cycle: 2, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

JWST is poised to revolutionize galaxy chemical abundances. First, it will enable accurate measurements of galaxy metallicities at high redshift using faint, temperature-sensitive emission lines and recalibration of the strong line method to high- z conditions. Second, it will qualitatively extend the redshift and mass ranges over which metallicities are measured. Scaling relations, which include the mass-metallicity relation (MZR) and the fundamental metallicity relation (FMR), encode extremely valuable information about fundamental galaxy evolution processes, including inflows and outflows. Analytic gas regulator models have been widely used to interpret the MZR and FMR. However, these models typically assume steady-state equilibrium conditions. In contrast, observations and simulations suggest that high- z and/or low-mass galaxies have highly time-variable (bursty) star formation rates (SFRs), inflows/outflows, and ISM conditions. The short time scales involved may strongly affect the predicted scaling relations and their scatter, and how chemical abundances relate to inflow and outflow properties. We will use high-resolution galaxy formation simulations to make new predictions for the MZR and FMR, with particular attention to regimes newly probed by JWST. The deliverables will include: (1) full predicted scaling relations from $z=0$ to $z\sim 10$; (2) quantification of scatter and deviations from standard relations, and how these correlate with regimes of bursty vs. steady star formation; (3) systematic tests of gas regulator models using realistic non-equilibrium simulations; (4)

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comparisons of predictions to available observations; and (5) inferred wind mass loading factors.

OBSERVING DESCRIPTION

This theory proposal will not make new observations.